

Strict pre-answer success prediction · the load-bearing quantity is the within-domain nested increment

Absolute AUROCs differ between domains by construction and are not comparable to each other; both nested increments exclude zero.

GSM8K

arithmetic word problems · 170 tasks / 680 candidates

chance = 0.50

Shortcut composite

0.731

Shortcut composite

Hidden alone

0.745

Hidden alone

Hidden + shortcuts

0.797

Hidden + shortcuts

paired $\Delta +0.066$

95% CI $[+0.021, +0.112]$ · excludes zero

components: length 0.687 · log-probability 0.569

0.5

0.6

0.7

0.8

AUROC

Horizon Logic

proof-depth-2-4 entailment · 201 heldout tasks / 614 scorable (pooled)

chance = 0.50

Shortcut composite

0.652

Hidden alone

0.763

Hidden + shortcuts

0.763

paired $\Delta +0.111$

95% CI $[+0.056, +0.169]$ · excludes zero

independent replication (510 new tasks): $+0.095 [+0.038, +0.157]$

robust to the adversarial malformed-sibling baseline ($+0.107$)

0.5

0.6

0.7

0.8

AUROC